EE3042/3014 Final Exam **Date: 17 January, 2023**

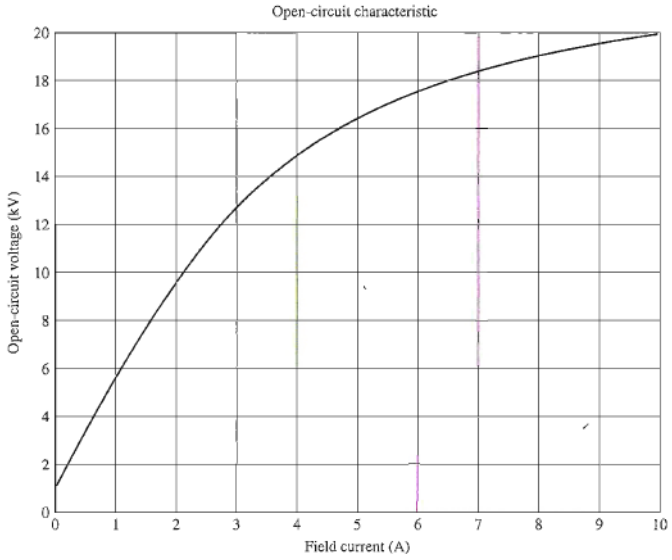
Student Name and Number _____

The notes and books will be kept closed. The calculator can be used. The necessary formulas are given at the last page.

1. (20p) Three-phase 20-kVA 24,000/277-V distribution transformer is connected in $\Delta - Y$. The open-circuit test was performed on the low-voltage side of this transformer bank, and the following data were recorded: $V_{line,OC} = 480\ V$, $I_{line,OC} = 4.05\ A$, $P_{3\emptyset,OC} = 948\ W$.
The short-circuit test was performed on the high-voltage side of this transformer bank, and the following data were recorded: $V_{line,SC} = 1400\ V$, $I_{line,SC} = 1.80\ A$, $P_{3\emptyset,SC} = 915\ W$
(a) Find the per-unit equivalent circuit of this transformer bank.
(b) Find the voltage regulation of this transformer bank at the rated load and 0.88 PF lagging.
(c) What is the transformer bank's efficiency under these conditions?

2. (20p) A 480-V, 102-kW, 0.82-pf-leading, 50-Hz, four-pole, Y-connected synchronous motor has a synchronous reactance of 1.8 Ohms and a negligible armature resistance. The rotational losses are also to be ignored. This motor is to be operated over a continuous range of speeds from 300 to 1500 r/min, where the speed changes are to be accomplished by controlling the system frequency with a solid-state drive.
(a) Over what range the input frequency must be varied to provide this speed control range?
(b) How large is E_a at the motor's rated conditions?
(c) What is the maximum power the motor can produce at rated speed with the E_a calculated in part (b)?

3. (20p) A 13.8-kV, 50-MVA, 0.9-power-factor-lagging, 60-Hz, four-pole wye-connected synchronous generator has a synchronous reactance of 2.5 Ohms and an armature resistance of 0.2 Ohm. Open circuit characteristic curve (OCC) for the generator is given as shown in the figure. Assume that the field current of the generator has been adjusted to value of 4.7A.
(a) What will the terminal voltage of this generator be if it is connected to a wye-connected load with an impedance of $30\angle 15$?
(b) Sketch the phasor diagram of this generator.



$P = E_A I_A$	$Q = V_\emptyset I_A sin\theta$ per phase	$P = \sqrt{3} V_T I_A cos\theta$	$P = \frac{3V_\emptyset E_A}{X_S} sin\delta$
$s = \frac{n_{sync} - n_m}{n_{sync}}$	$SR = \frac{n_{nl} - n_{fl}}{n_{fl}}$	$P_{conv} = (1-s)P_{AG}$ $= P_{AG} - P_{RCL}$	$f_r = sf_e$
$E_A = K\emptyset w$	$Pout = w\,\tau$	$P = E_A I_A$	
$for\ sysnc\ motor\ E_A$ $= V_\emptyset - R_A I_A - jX_s I_A$	$for\ sysnc\ gen\ \ E_A = V_\emptyset + R_A I_A + jX_s I_A$	$Z_{base} = \frac{V_{\emptyset,\text{base}}}{S_{\emptyset,\text{base}}}$	$Y_{EX} = G_C - jB_M$

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Q.1) (35p) 20-kVA 24,000/277-V distribution TR, Δ-Y connected

(LV side) open loop test: $V_{line,oc} = 480V$, $I_{line,oc} = 4.05A$, $P_{3\phi,oc} = 948W$

(HV side) The short circuit test: $V_{line,sc} = 1400V$, $I_{line,sc} = 1,8A$, $P_{3\phi,sc} = 915W$

a) per unit equivalent circuit, b) VR? at rated load and 0,8 PF lagging

c) $\eta_{TR} = ?$

a) from the OC test (Y → low voltage side)

$$|Y_{EX}| = \frac{I_{\phi,oc}}{V_{\phi,oc}} = \frac{4,05A}{480/\sqrt{3}\,V} = 0,0146\,S \quad 1/$$

$$Y_{EX} = \frac{I_{\phi,oc}}{V_{\phi,oc}}\angle -\theta \Rightarrow \theta = \cos^{-1}\left(\frac{P_{\phi,oc}}{(V_{\phi,oc})(I_{\phi,oc})}\right) = \cos^{-1}\left(\frac{948/3\ W}{(480/\sqrt{3})(4.05)}\right) = +73.65° \quad 1/$$

$$Y_{EX} = 0,0146\angle -73.65° = 4,0824\times10^{-3} - j0,0139 = G_c - jB_M \quad 1/$$

$$R_c = \frac{1}{G_c} = 244.95\Omega \quad 1/, \quad X_M = \frac{1}{B_M} = 71,94\Omega \quad 1/$$

Zbase referred to the LVS (Low-voltage side, secondary): $Z_{base,s} = \frac{(V_{\phi,base,s})^2}{S_{\phi,base}} = \frac{(277/\sqrt{3})^2}{20kVA/3}$

$$Z_{base,s} = 3.84\Omega \quad 1/$$

From the SC test (HVS, Δ)

$$|Z_{EQ}| = \frac{V_{\phi,sc}}{I_{\phi,sc}} = \frac{1400}{1,8/\sqrt{3}} = 1347,15\Omega \quad 1/$$

$$Z_{EQ} = \frac{V_{\phi,sc}}{I_{\phi,sc}}\angle\theta \Rightarrow \theta = \cos^{-1}\left(\frac{P_{\phi,sc}}{V_{\phi,sc}\,I_{\phi,sc}}\right) = \cos^{-1}\underbrace{\left(\frac{915/3}{(1400)(1,8/\sqrt{3})}\right)}_{0,21} = 77.9° \quad 1/$$

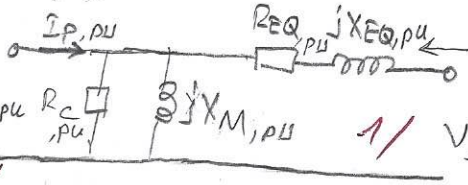
$$Z_{EQ} = R_{EQ,P} + jX_{EX,P} = 1347.15\angle 77.9°\,\Omega = 282,4 + j1317,22\,\Omega \quad 1/$$

P: primary
S: secondary
PU: per unit

$$Z_{base,P} = \frac{(V_{\phi,base,P})^2}{S_{\phi,base}} = \frac{(24000V)^2}{20kVA/3} \Rightarrow Z_{base,P} = 86,4K\Omega \quad 1/$$

$$R_{EQ,pu} = \frac{R_{EQ,P}}{Z_{base,P}} = \frac{282.4\Omega}{86.4K\Omega} = 3,27\times10^{-3}\ pu \quad 1/$$

$$X_{EQ,pu} = \frac{X_{EQ,P}}{Z_{base,P}} = \frac{1317.22\Omega}{86.4K\Omega} = 15.25\times10^{-3}\ pu \quad 1/$$



$R_{c,pu} = 244.95/3,84 = 63,79$ 1/, $X_{M,pu} = 71,94/3,84 = 18.73 \Leftarrow (LVS)$ 1/

Q1. continued

b) $PF = \cos\theta = 0{,}88 \Rightarrow \theta = \cos^{-1}(0{,}88) = 28{,}36°$ lagging at rated load 1/

/phase $\Rightarrow V_{P,pu} = V_{S,pu} + (Z_{EQ,P,pu})\, I_{S,pu} \Rightarrow V_{S,pu} = \frac{277/\sqrt{3}}{277/\sqrt{3}} = 1$ pu 1/

$V_{\phi,base}$

$S_{base,P} = S_{base,S} = 20$kVA 1/

$I_S = \frac{S_{\phi,base}}{V_{\phi,S,base}} = \frac{20\text{kVA}/3}{277/\sqrt{3}} = 41.69$ A 2/

$V_{P,pu} = 1\angle 0° + \frac{(3{,}27 + j15{,}25)\times 10^{-3}}{0{,}01556\angle 77.9°}\,(1\angle -28{,}36)$

$I_{S,pu} = \frac{I_{S,base}}{I_{S,base}} = 1$ pu 1/

$V_{P,PU} = 1{,}01\angle 0{,}67°$ 2/

$VR = \frac{1.01 - 1.0}{1.0} \times 100 = 1\%$ 2/

c) $P_{out\,(pu)} = 3 V_{S,pu} I_{S,pu} \cos\theta = (3)(1{,}0)(1{,}0)(0{,}88) = 2{,}64$ pu 2/

$P_{cu,pu} = 3 I_{S,pu}^2 R_{EQ,pu} = (3)(1.0)^2 (3{,}27\times 10^{-3}) = 9{,}81\times 10^{-3}$ pu 2/

$P_{core,pu} = 3\frac{V_{P,cu}^2}{R_{c,pu}} = 3\frac{(1.01)^2}{63{,}79} = 0{,}048$ pu 2/

$P_{in,pu} = P_{out,pu} + P_{cu,pu} + P_{core,pu} = 2.64 + 9{,}81\times 10^{-3} + 0{,}048 \Rightarrow$

$P_{in,pu} = 2.7$ pu 2/

$\eta = \frac{P_{out,pu}}{P_{in,pu}} \times 100 = \frac{2.64}{2{,}7} \times 100 = 97{,}78\%$ 2/

The efficiency based on power/phase also yields the same result.

Q.2) (35p) A 480-V, 102-kW, 0,82-pf-loading, 4P, Y-connected

SM $X_S = 1{,}8\,\Omega$, $R_A \cong 0$, $P_{rot} \cong 0$, Range of speed [300-1500] r/m.

a) f range? → [300 − 1500] r/m.

b) $E_{A,rated}$?

c) P_{max} ? at rated speed with E_A calculated in b)

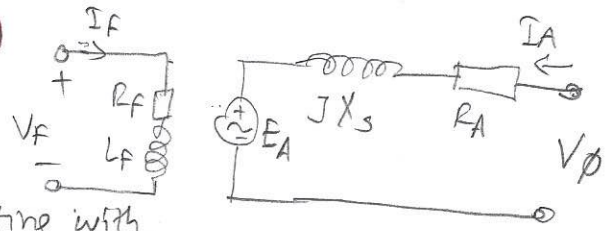
Q2-continued

a) $f_{se} = \frac{P n_m}{120} \Rightarrow n_m = 300 \text{ r/min} \Rightarrow f_{se(300)} = \frac{(4)(300)}{120} = 10 \text{Hz}$. 5/

$n_m = 1500 \text{ r/min} \Rightarrow f_{se(1500)} = \frac{(4)(1500)}{120} = 50 \text{Hz}$. 5/

The frequency (controlled) range ise 10Hz - 50Hz.

b)



rotating with the synchronous speed.

$E_A = V\phi - (R_A + jX_s) I_A$

$E_A = V\phi - X_s I_A \angle +\theta + 90°$ (rated) ⇓

5/ $I_\phi = I_A = I_L = \frac{P}{\sqrt{3} V_T \cos\theta} = \frac{102 \text{kW}}{\sqrt{3}(480)(0{,}82)} = 149.6 \text{ A}$

$I_A = 149.6 \angle 34.92°$

5/ $V\phi = \frac{480 \text{V}}{\sqrt{3}} = 277.13 \angle 0° \text{ V}$, $\theta = \cos^{-1}(0{,}82) = 34.92°$

$E_A = 277.13 \angle 0° - \underbrace{(1{,}8)(149.6)}_{269.28} \angle +34.92+90 = 484.51 \angle -27.11° \text{ V}$ 10/

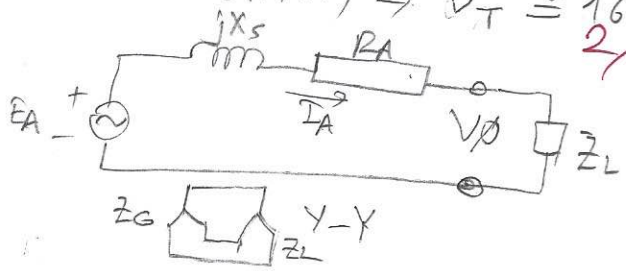
c) $P_{max} = \frac{3 V\phi E_A}{X_s} = \frac{3(277)(484.51)}{1{,}8} = 223.68 \text{ kW}$ 5/

Q.3 (30p) A 13,8-kV, 50-MVA, 0,9-pf-loging, 60-Hz, 4-pole Y-connected SG ⇒ $X_s = 2.5\Omega$, $R_A = 0{,}2\Omega$, $I_F = 4.7\text{A}$ (adjusted)

a) V_T =? Y-connected load ⇒ $Z_L = 30 \angle 15°$ b) phasor diagram?

with.

a) From the OCC curve, the corresponding E_A for the adjusted I_F (4.7A) ⇒ $V_T \cong 16\text{kV}$ ⇒ $E_A = V_T/\sqrt{3} = 9.24 \text{ kV}$ (Y-connected) 2/



$E_A \angle \delta = (Z_s + Z_L) I_A$

$\Rightarrow Z_s = 0{,}2 + j2.5 = 2.51 \angle 85.4$, $Z_L = 30 \angle 15°$ 2/

$\frac{16\text{kV}}{\sqrt{3}} \angle \delta = (2.51 \angle 85.43 + 30 \angle 15°) I_A \angle -\theta$ 3/

$I_A \angle -\theta = \frac{16\text{kV}/\sqrt{3} \angle 0°}{31 \angle 19.39°} = 298 \angle -19.39° \text{ kV}$, $\theta = 19.39°$ 2/

$\delta = 0$ (taking as a reference 6/

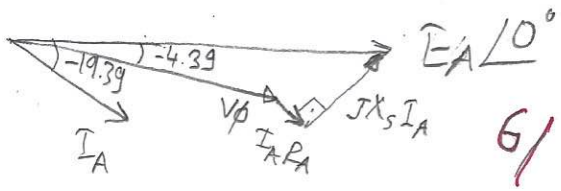
$V\phi = Z_L I_A$ or $V\phi = E_A \angle 0° - (R_A + jX_s) I_A$

$E_A \cong 9.24 \angle 0° \text{ kV}$ 8/

Q.3 - continued

$$V_\phi = (30\angle 15)(298\angle -19.39) = 8940\angle -4.39$$ 3/

b)



6/

$V_T = \sqrt{3}\, V_\phi$

$= (\sqrt{3})(8940)$

$\cong 15.5\angle -4.39°\ kV$

1/

4/4